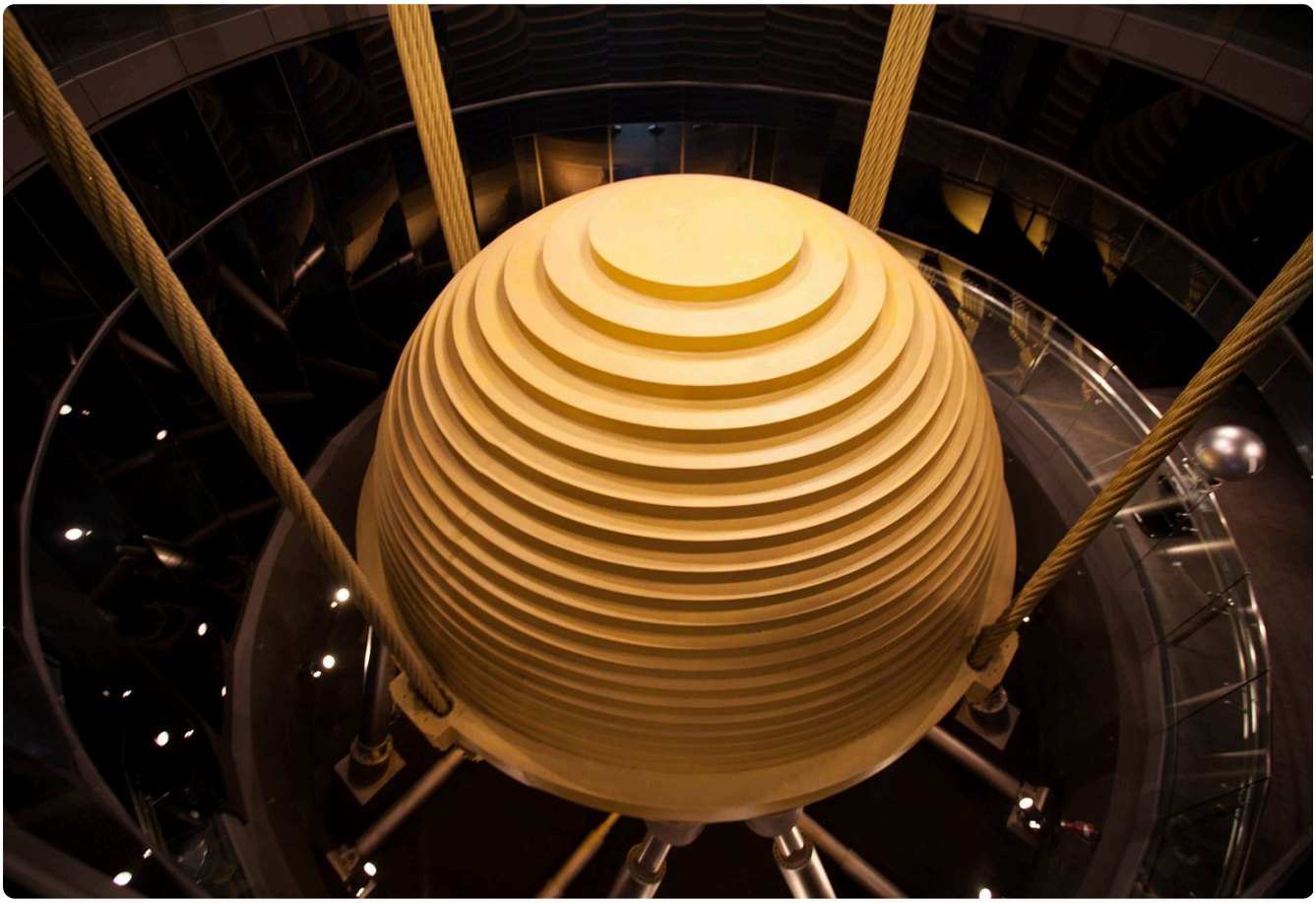
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Fight motion with motion: the tuned mass damper

The free guide to the reel on Taipei 101: how a 660-tonne steel ball keeps a 508 m tower steady, how engineers tune it, and how to build a working one on your desk.



How it works

Every tall building sways a little, at its own natural rhythm, like a ruler you flick on the edge of a table. When gusts or an earthquake push it in time with that rhythm, the movement grows. Making the whole tower stiffer reduces it, but it costs a great deal of steel and concrete.

A tuned mass damper takes the other route. A heavy mass hangs or slides near the top of the building and is tuned to swing at the same rhythm as the tower. When the tower leans one way, the mass lags behind and pulls the other way. Hydraulic dampers (big shock absorbers) connect the two and turn the motion into heat.

The key word is tuned: a pendulum's swing time depends only on its length, so engineers choose the cable length that matches the building's sway period, then adjust it on site after measuring the real tower.

Taipei 101 in numbers

Taipei 101 is 508 m tall and was the world's tallest building from 2004 to 2010. Its damper is a sphere 5.5 m wide made of 41 steel plates, each 125 mm thick, about 660 tonnes in all. It hangs on cables from the 92nd floor down to the 87th, where visitors can see it from the observatory.

The tower's operator says it cuts sway by up to 40 %. Its record movement was 1 m during Typhoon Soudelor in August 2015, short of the 1.5 m it was designed to allow (Guinness World Records lists it as the largest tuned mass damper).

Tuning it: the one formula you need

A simple pendulum swings back and forth in a time $T = 2\pi \times \sqrt{(L / g)}$, where L is the length of the cable and $g = 9.81\ m/s^2$. Turn it around to find the length: $L = g \times (T / 2\pi)^2$.

Example: if a tower sways back and forth once every 7 seconds, $L = 9.81 \times (7 / 6.28)^2 \approx 12$ m of cable. A shorter cable swings faster, a longer one slower. Real dampers also tune the damping (how much energy the shock absorbers take) so the mass does not just swing forever.

Built examples

Taipei 101, Taiwan (2004)	A 660-tonne steel sphere on cables between floors 87 and 92, with hydraulic dampers; open to visitors.
Citicorp Center (now 601 Lexington), New York (1977)	One of the first skyscrapers with a tuned mass damper: a block of about 400 tonnes of concrete that slides on a thin film of oil near the top.
Shanghai Tower, China (2015)	A damper of about 1,000 tonnes that uses magnets (eddy currents) instead of oil to absorb the motion.
Millennium Bridge, London (2002)	After it wobbled under walkers on opening day in 2000, it was closed and fitted with dampers, including tuned mass dampers under the deck.

Try it yourself: a desk tower with a damper

1. Build a tall, thin tower from a strip of thin plywood or stiff cardboard (about 60 cm high) fixed upright in a heavy base.
2. Flick the top and count how many full back-and-forth swings it makes in 10 seconds. Divide 10 by that number: this is the tower's period T.
3. Compute the pendulum length $L = 9.81 \times (T / 6.28)^2$ in metres. A period of 0.5 s gives about 6 cm.
4. Hang a small weight (a metal nut or a few coins in tape) on a thread of that length from a little arm at the top of the tower.
5. Flick the tower again with and without the pendulum and film both with your phone: with the right length, the swaying dies out much faster.
6. Change the thread length by 2 cm longer or shorter and repeat: you will see the effect weaken. That is what tuned means.

The rule to remember

If something moves, you do not always need to make it heavier or stiffer. Add a mass that moves against it, tuned to the same rhythm.

Brute force	Thicker walls and more steel everywhere: works, but costs material and money.
Untuned mass	A weight that swings at the wrong speed does almost nothing.
No damping	Without shock absorbers the mass keeps swinging; the energy has to go somewhere.

Go further

Tuned mass damper (Wikipedia)
https://en.wikipedia.org/wiki/Tuned_mass_damper

Largest tuned mass damper (Guinness World Records)
https://www.guinnessworldrecords.com/world-records/774616-largest-tuned-mass-damper

Taipei 101 (Wikipedia)
https://en.wikipedia.org/wiki/Taipei_101

Credits
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Facts: Guinness World Records, Taiwan News, Taipei 101 Observatory, Wikipedia (Citicorp Center, Shanghai Tower, Millennium Bridge).

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